

Auteur: Siebe Haché
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 46.1

$$\int \frac{8(x^3 - 1)}{9x^4 + 10x^2 + 1} dx$$

Ontbinding noemer:

$$\text{Stel } y = x^2 \Rightarrow 9y^2 + 10y + 1 = 0.$$

$$D = 10^2 - 4 \cdot 9 \cdot 1 = 100 - 36 = 64.$$

$$y = \frac{-10 \pm 8}{18} \Rightarrow y_1 = -1 \text{ en } y_2 = -\frac{1}{9}.$$

$$\text{Dus de noemer is: } 9(x^2 + 1) \left(x^2 + \frac{1}{9}\right)$$

$$\int \frac{8(x^3 - 1)}{9x^4 + 10x^2 + 1} dx = \frac{8}{9} \int \frac{x^3 - 1}{(x^2 + 1) \left(x^2 + \frac{1}{9}\right)} dx$$

$$\Rightarrow \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + \frac{1}{9}}$$

$$A(i) + B = \frac{i^3 - 1}{-1 + \frac{1}{9}} = \frac{-i - 1}{-\frac{8}{9}} = \frac{9}{8} + \frac{9}{8}i$$

$$\Rightarrow A = \frac{9}{8} \text{ en } B = \frac{9}{8}$$

$$C\left(\frac{1}{3}i\right) + D = \frac{\left(\frac{1}{3}i\right)^3 - 1}{-\frac{1}{9} + 1} = \frac{-\frac{1}{27}i - 1}{\frac{8}{9}} = -\frac{9}{8} - \frac{1}{24}i$$

$$\Rightarrow C = -\frac{1}{8} \text{ en } D = -\frac{9}{8}$$

$$= \frac{8}{9} \left(\int \frac{\frac{9}{8}x + \frac{9}{8}}{x^2 + 1} dx + \int \frac{-\frac{1}{8}x - \frac{9}{8}}{x^2 + \frac{1}{9}} dx \right)$$

$$= \int \frac{x + 1}{x^2 + 1} dx - \frac{1}{9} \int \frac{x + 9}{x^2 + \frac{1}{9}} dx$$

$$= \int \frac{1}{x^2 + 1} dx + \frac{1}{2} \int \frac{1}{x^2 + 1} d(x^2 + 1) - \frac{1}{18} \int \frac{1}{x^2 + \frac{1}{9}} d\left(x^2 + \frac{1}{9}\right) - \int \frac{1}{x^2 + \frac{1}{9}} dx$$

$$= \text{Bgtan}(x) + \frac{1}{2} \ln|x^2 + 1| - \frac{1}{18} \ln\left|x^2 + \frac{1}{9}\right| - 3 \text{Bgtan}(3x) + C$$

References